

Indian Institute of Technology Gandhinagar

ME 643 — Aircraft and Rocket Propulsion

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## Aircraft Engine Design Project

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**Aircraft:** Gulfstream G650

**Engine:** Rolls-Royce BR725-A1-12 (Turbofan)

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Group 8 — 2026

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## 1 Introduction

The selected mission route is **Sardar Vallabhbhai Patel International Airport (AMD), Ahmedabad to Heathrow Airport (LHR), London**, covering a great-circle distance of approximately 6,900 km.

The primary objectives are: (1) establish aircraft and journey specifications via an RFP; (2) conduct phase-by-phase mission analysis to compute thrust requirements, fuel burn, and weight evolution; (3) perform Parametric Cycle Analysis (PCA) to determine optimal engine design parameters; and (4) carry out Engine Performance Analysis (EPA) using PERF to verify that the design meets all mission requirements including the OEI climb gradient constraint.

## 2 Request for Proposal (RFP)

### 2.1 Aircraft and Journey Specifications

**Table 1.** Aircraft specifications — Gulfstream G650

| Parameter                 | Symbol / Unit              | Value                                |
|---------------------------|----------------------------|--------------------------------------|
| Seating capacity          | —                          | 13–16                                |
| Maximum payload           | $W_{\text{payload}}$ (N)   | 28,900                               |
| Maximum fuel capacity     | $m_{\text{fuel,max}}$ (kg) | 20,049                               |
| Maximum take-off weight   | MTOW (kg)                  | 45,178                               |
| Maximum landing weight    | $W_{\text{land}}$ (kg)     | 37,875                               |
| Wing area                 | $S$ (m <sup>2</sup> )      | 119.2                                |
| Aspect ratio              | $AR$                       | 7.75                                 |
| Engine type               | —                          | Turbofan $\times 2$ (RR BR725-A1-12) |
| Max engine inlet diameter | $D$ (m)                    | 1.27                                 |
| Maximum lift coefficient  | $C_{L,\text{max}}$         | 1.6                                  |
| Turbine inlet temperature | $T_{t4}$ (K)               | 1500                                 |
| Calorific value (Jet-A)   | $h_{PR}$ (MJ/kg)           | 43                                   |

**Table 2.** Journey specifications — AMD to LHR

| Parameter                    | Value                                                     |
|------------------------------|-----------------------------------------------------------|
| Route                        | AMD (Ahmedabad) $\rightarrow$ LHR (London)                |
| Total range                  | 6,900 km                                                  |
| Cruise altitude / Mach       | 15.5 km / $M_0 = 0.85$                                    |
| Runway length (AMD)          | 3,500 m (sea level, $T_{\text{amb}} = 38^\circ\text{C}$ ) |
| Single-engine climb gradient | 2.4%                                                      |

## 3 Mission Analysis

### 3.1 Mission Profile and Assumed TSFC

An initial average TSFC of 0.250 kg/(N · hr) is assumed, consistent with published BR725 cruise performance:

$$\text{TSFC} = \frac{0.250}{3600} = 6.944 \times 10^{-5} \text{ kg}/(\text{N} \cdot \text{s})$$

Climb/descent distances are derived from the OEI gradient ( $\sin \theta = 0.024$ ):

$$d_{\text{climb}} = d_{\text{descent}} = \frac{15,500}{0.024} = 645.8 \text{ km}$$

**Table 3.** Phase-wise distance breakdown

| Phase          | Distance (km) | Basis                             |
|----------------|---------------|-----------------------------------|
| Taxi (out)     | 5.0           | Assumed                           |
| Takeoff        | 20.0          | Assumed                           |
| Climb          | 645.8         | $h_{\text{cruise}} / \sin \theta$ |
| Cruise         | 5,558.4       | Remainder                         |
| Descent        | 645.8         | $h_{\text{cruise}} / \sin \theta$ |
| Landing & Taxi | 25.0          | Assumed                           |
| <b>Total</b>   | <b>6,900</b>  |                                   |

### 3.2 Phase-by-Phase Calculations

#### 3.2.1 Taxi (Out)

Ground speed 10 m/s; idle thrust 7.5 kN.

$$t_{\text{taxi}} = 500 \text{ s}; \quad F_{\text{taxi}} = 260 \text{ kg}; \quad W_{\text{end}} = 44,918 \text{ kg}$$

#### 3.2.2 Takeoff

Sea level;  $C_{L,\text{max}} = 1.6$ ;  $T/W = 0.3$ .

$$V_{\text{stall}} = \sqrt{\frac{2W}{\rho S C_{L,\text{max}}}} = 63.84 \text{ m/s}; \quad V_{\text{TO}} = 1.2 V_{\text{stall}} = 77 \text{ m/s} \quad (1)$$

$$C_L = 1.099; \quad C_{D,i} = \frac{C_L^2}{\pi A R e} = 0.0584; \quad T = 132.2 \text{ kN} \quad (2)$$

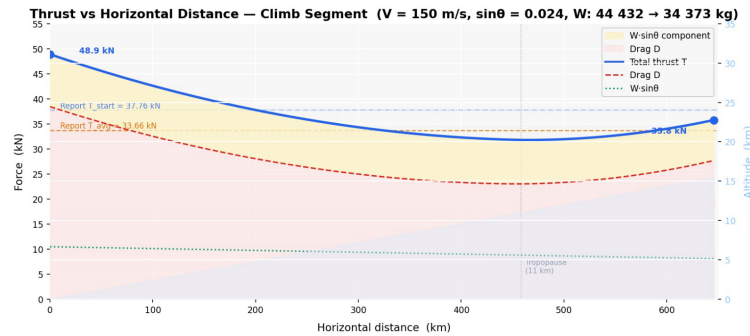
$$t_{\text{TO}} = 53 \text{ s}; \quad F_{\text{TO}} = 486 \text{ kg}; \quad W_{\text{end}} = 44,432 \text{ kg}$$

#### 3.2.3 Climb

Average speed 150 m/s;  $\rho_{\text{avg}} = 0.568 \text{ kg/m}^3$ ;  $t_{\text{climb}} = 4,306 \text{ s}$ .

$$T_{\text{climb,avg}} = D_{\text{mid}} + W_{\text{mid}} \sin \theta = 24487 + 9169 = 33.7 \text{ kN}$$

$$F_{\text{climb}} = 10,059 \text{ kg}; \quad W_{\text{end}} = 34,373 \text{ kg}$$



**Figure 1.** Climb performance plot: Thrust vs. horizontal distance.

#### 3.2.4 Cruise

$M_0 = 0.85$ ;  $V_c = 250.75 \text{ m/s}$ ;  $t_{\text{cruise}} = 22,165 \text{ s}$ .

$$C_L = 0.482; \quad C_D = 0.03124; \quad T = D = 21.8 \text{ kN} \quad (3)$$

$$F_{\text{cruise}} = 3,359 \text{ kg}; \quad W_{\text{end}} = 31,014 \text{ kg} \quad (4)$$

### 3.2.5 Descent

Average speed 180 m/s; near-idle thrust 5.0 kN.

$$t_{\text{descent}} = 3,588 \text{ s}; \quad F_{\text{descent}} = 1,246 \text{ kg}; \quad W_{\text{end}} = 29,768 \text{ kg}$$

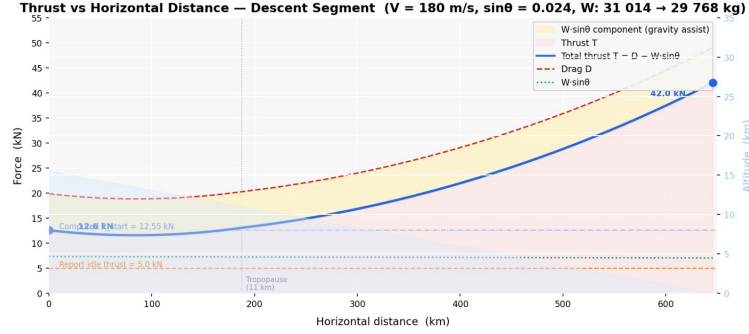


Figure 2. Descent performance plot: Thrust vs. horizontal distance.

### 3.2.6 Landing and Taxi (In)

$$t_{\text{land}} = 625 \text{ s}; \quad F_{\text{land}} = 325 \text{ kg}; \quad W_{\text{final}} = 29,443 \text{ kg}$$

## 3.3 Mission Summary

Table 4. Thrust requirement, fuel burn, and weight variation across mission phases

| Phase        | Dist.(km)   | Thrust(kN) | Time(s)      | Fuel(kg)     | $W_{\text{start}}$ (kg) | $W_{\text{end}}$ (kg) |
|--------------|-------------|------------|--------------|--------------|-------------------------|-----------------------|
| Taxi (out)   | 5.0         | 7.5        | 500          | 260          | 45178                   | 44918                 |
| Takeoff      | 20.0        | 132.2      | 53           | 486          | 44918                   | 44432                 |
| Climb        | 645.8       | 33.7       | 4306         | 10059        | 44432                   | 34373                 |
| Cruise       | 5558.4      | 21.8       | 22165        | 3359         | 34373                   | 31014                 |
| Descent      | 645.8       | 5.0        | 3588         | 1246         | 31014                   | 29768                 |
| Land & Taxi  | 25.0        | 7.5        | 625          | 325          | 29768                   | 29443                 |
| <b>Total</b> | <b>6900</b> | —          | <b>31237</b> | <b>15735</b> | 45178                   | 29443                 |

Total fuel burn: 15,735 kg. Fuel reserve =  $20049 - 15735 = 4,314$  kg (21.5% margin, above the 10% minimum). The climb phase dominates ( $\approx 64\%$  of total fuel) due to the OEL-gradient-derived 645.8 km climb at near full thrust.

## 4 Parametric Cycle Analysis (PCA)

### 4.1 Fixed Design Point Parameters

All cycle calculations are performed at the **cruise design point** (Table 5).

**Table 5.** Fixed design point parameters for PCA

| Parameter               | Symbol             | Value                     |                                |
|-------------------------|--------------------|---------------------------|--------------------------------|
| Cruise Mach number      | $M_0$              | 0.85                      |                                |
| Ambient temperature     | $T_0$              | 216.65 K                  | ISA @ 15.5 km                  |
| Ambient pressure        | $P_0$              | 11,597 Pa                 | ISA                            |
| Air density             | $\rho_0$           | 0.186,5 kg/m <sup>3</sup> | ISA                            |
| Ratio of specific heats | $\gamma$           | 1.4                       | Assumed                        |
| Specific heat           | $c_p$              | 1,004 J/(kg · K)          | Assumed                        |
| Fuel heating value      | $h_{PR}$           | 43,000 kJ/kg              |                                |
| Turbine inlet temp.     | $T_{t4}$           | 1,500 K                   |                                |
| Max inlet diameter      | $D$                | 1.27 m                    |                                |
| Inlet capture area      | $A_{\text{inlet}}$ | 1.267 m <sup>2</sup>      | $\pi D^2/4$                    |
| Speed of sound          | $a_0$              | 295.0 m/s                 | $\sqrt{\gamma R T_0}$          |
| Ram temperature ratio   | $\tau_r$           | 1.1445                    | $1 + \frac{\gamma-1}{2} M_0^2$ |
| Thermal limit param.    | $\tau_\lambda$     | 6.924                     | $T_{t4}/T_0$                   |

## 4.2 Design Point and Constraints

PCA is performed at cruise ( $M_0 = 0.85$ ,  $h = 15.5$  km,  $T_0 = 216.65$  K,  $T_{t4} = 1,500$  K). Key derived parameters:  $\tau_r = 1.1445$ ,  $\tau_\lambda = 6.924$ .

**Mission constraints:** Maximum TSFC  $S_{\text{max}} = 69.4$  mg/(N · s); minimum specific thrust:

$$A_{\text{inlet}} = \frac{\pi(1.27)^2}{4} = 1.267 \text{ m}^2; \quad \dot{m}_{0,\text{max}} = 0.1865 \times 250.7 \times 1.267 = 59.2 \text{ kg/s}; \quad ST_{\text{min}} = \frac{10913}{59.2} = 184 \text{ N/(kg/s)}$$

## 4.3 Governing Equations — Ideal Turbofan

$$\tau_c = \pi_c^{2/7}, \quad \tau_f = \pi_f^{2/7}, \quad \tau_t = 1 - \frac{\tau_r}{\tau_\lambda} [\tau_c - 1 + \alpha(\tau_f - 1)] \quad (5)$$

$$\left(\frac{V_9}{a_0}\right)^2 = \frac{2}{\gamma - 1} \left[ \tau_\lambda - \tau_r(\tau_c - 1 + \alpha(\tau_f - 1)) - \frac{\tau_\lambda}{\tau_r \tau_c} \right]; \quad \left(\frac{V_{19}}{a_0}\right)^2 = \frac{2}{\gamma - 1} [\tau_r \tau_f - 1] \quad (6)$$

$$\frac{F}{\dot{m}_0} = \frac{a_0}{1 + \alpha} \left[ \frac{V_9}{a_0} - M_0 + \alpha \left( \frac{V_{19}}{a_0} - M_0 \right) \right]; \quad f = \frac{c_p T_0}{h_{PR}} (\tau_\lambda - \tau_r \tau_c); \quad S = \frac{f}{(1 + \alpha)(F/\dot{m}_0)} \quad (7)$$

The parametric ranges explored:  $\pi_c \in [10, 40]$ ,  $\pi_f \in [1.1, 2.5]$ ,  $\alpha \in [1, 12]$ . Feasibility requires  $\tau_T > 0$ ,  $(V_9/a_0)^2 > 0$ , and  $f > 0$ . At  $T_{t4} = 1,500$  K, no design with  $\alpha > 7.0$  can satisfy  $ST_{\text{min}}$ ; the feasible range is  $\alpha \in [1, 7]$ .

## 4.4 Variable Range Justification

Three feasibility conditions:

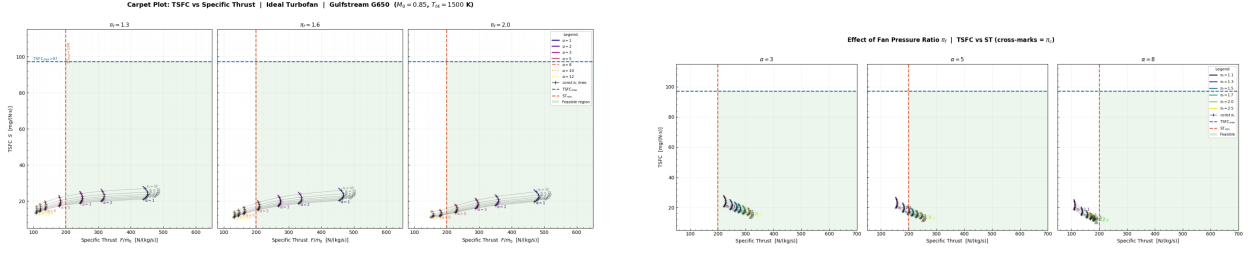
- (i)  $\tau_t > 0$  (turbine not thermodynamically over-expanded),
- (ii)  $(V_9/a_0)^2 > 0$  (core nozzle physically feasible),
- (iii)  $f > 0$  (fuel is added, not removed).

**Compressor pressure ratio**  $\pi_c \in [10, 40]$ : At  $\pi_c = 10$ :  $\tau_c = 10^{2/7} = 2.069$ , giving  $\eta_T = 1 - 1/(1.1445 \times 2.069) = 57.9\%$  — marginal but acceptable. The upper bound  $\pi_c = 40$  is the practical limit by compressor blade aerodynamics and material strength at  $T_{t4} = 1,500$  K and  $\pi_c \lesssim 125$ .

**Fan pressure ratio**  $\pi_f \in [1.1, 2.5]$ :  $\pi_f = 1$  degenerates to a turbojet. The practical upper limit for a subsonic fan is  $\pi_f \lesssim 2.0$ – $2.2$  (fan tip relative Mach number exceeds unity beyond this, incurring shock losses).

**Bypass ratio  $\alpha \in [1, 12]$ :**  $\alpha = 0$  is a turbojet. A systematic scan reveals that **no design with  $\alpha > 7.0$  can satisfy  $ST_{\min} = 199 \text{ N}/(\text{kg/s})$  at  $T_{t4} = 1,500 \text{ K}$  and the given inlet geometry.** This is a fundamental thermodynamic constraint, not a manufacturing limit.

#### 4.5 Carpet Plots and Design Selection



(a) TSFC vs  $ST$ : effect of  $\alpha$  and  $\pi_c$  for  $\pi_f = 1.3, 1.6, 2.0$ . (b) Effect of  $\pi_f$  for  $\alpha = 3, 5, 8$ ; cross-marks = const.  $\pi_c$ . **Figure 3.** PCA carpet plots. **Blue line:**  $S_{\max} = 69.4 \text{ mg}/(\text{N} \cdot \text{s})$ ; **Red line:**  $ST_{\min} = 184 \text{ N}/(\text{kg/s})$ . Green region = feasible.

**Table 6.** PCA design options — performance at cruise design point

| Option                     | $\alpha$ | $\pi_c$   | $\pi_f$    | $F/\dot{m}_0$ [N/(kg/s)] | $S$ [mg/(N · s)] | $\eta_T$     | $\eta_P$     | $\eta_O$     |
|----------------------------|----------|-----------|------------|--------------------------|------------------|--------------|--------------|--------------|
| A — Fuel Economy           | 5        | 40        | 1.8        | 217                      | 14.2             | 69.5%        | 59.2%        | 41.1%        |
| <b>B — Balanced (rec.)</b> | <b>4</b> | <b>30</b> | <b>1.7</b> | <b>240</b>               | <b>16.4</b>      | <b>66.9%</b> | <b>53.0%</b> | <b>35.5%</b> |
| C — Thrust Margin          | 3        | 25        | 1.8        | 286                      | 17.9             | 65.2%        | 49.9%        | 32.5%        |
| Constraint                 | —        | —         | —          | $> 184$                  | $< 69.4$         | —            | —            | —            |

**Option B** ( $\alpha = 4$ ,  $\pi_c = 30$ ,  $\pi_f = 1.7$ ) is recommended: it provides a 30%  $ST$  margin above  $ST_{\min}$ , uses mature  $\pi_c = 30$  technology, and achieves  $\eta_O = 35.5\%$ . Option A has the best TSFC but only a 9%  $ST$  margin, while Option C gives maximum thrust headroom at the cost of efficiency.

## 5 Engine Performance Analysis (EPA)

### 5.1 Reference Engine Setup (PERF V5.0)

Option B parameters were entered into PERF V5.0 (Turbofan — 2 Exhausts).  $T_{t4}$  was raised to 1,750 K for the non-ideal cycle. Reference mass flow: 50 kg/s per engine.

**Table 7.** Reference engine inputs and component efficiencies

| Parameter     | Symbol      | Value   | Component     | Efficiency   | Value  |
|---------------|-------------|---------|---------------|--------------|--------|
| Compressor PR | $\pi_c$     | 30      | LP Compressor | $\eta_{LPC}$ | 88.15% |
| Fan PR        | $\pi_f$     | 1.7     | HP Compressor | $\eta_{HPC}$ | 85.43% |
| LPC PR        | $\pi_{LPC}$ | 1.7     | Fan           | $\eta_{fan}$ | 88.15% |
| Bypass ratio  | $\alpha$    | 4.0     | HP Turbine    | $\eta_{HPT}$ | 90.17% |
| $T_{t4}$      | —           | 1750 K  | LP Turbine    | $\eta_{LPT}$ | 90.68% |
| Mass flow     | $\dot{m}_0$ | 50 kg/s | —             | —            | —      |

## 5.2 Reference Engine at Cruise

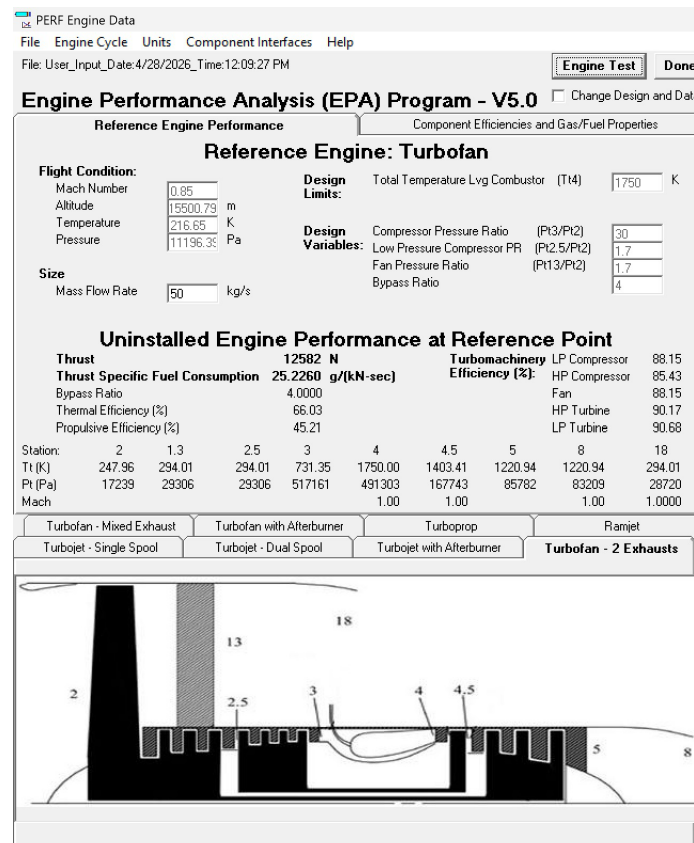


Figure 4. PERF output at cruise:  $M_0 = 0.85$ , altitude = 15,500 m.

Key cruise results: thrust/engine = 12,582 N (total 25,164 N vs. required 21,826 N, +15.3%); TSFC = 25.226 g/(kN · s);  $\eta_T = 66.03\%$ ;  $\eta_P = 45.21\%$ ;  $T_{t3} = 731$  K,  $T_{t5} = 1,221$  K.

### 5.3 Engine Tests at Mission Journey Points

#### 5.3.1 Takeoff — $M_0 = 0.2$ , Sea Level

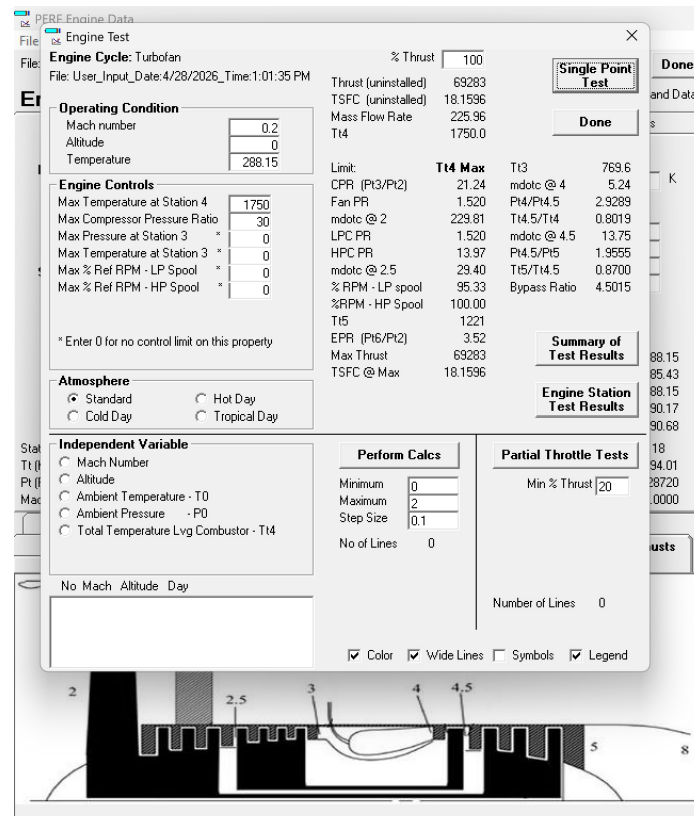


Figure 5. PERF result at takeoff:  $M_0 = 0.2$ , Alt = 0 m.

Thrust available = 69,283 N/engine vs. required 66,100 N  $\Rightarrow$  +4.8% **margin**. TSFC = 18.16 g/(kN · s);  $\dot{m}_0 = 225.96$  kg/s; bypass ratio = 4.50.

### 5.3.2 Climb — $M_0 = 0.65$ , Altitude = 9,000 m

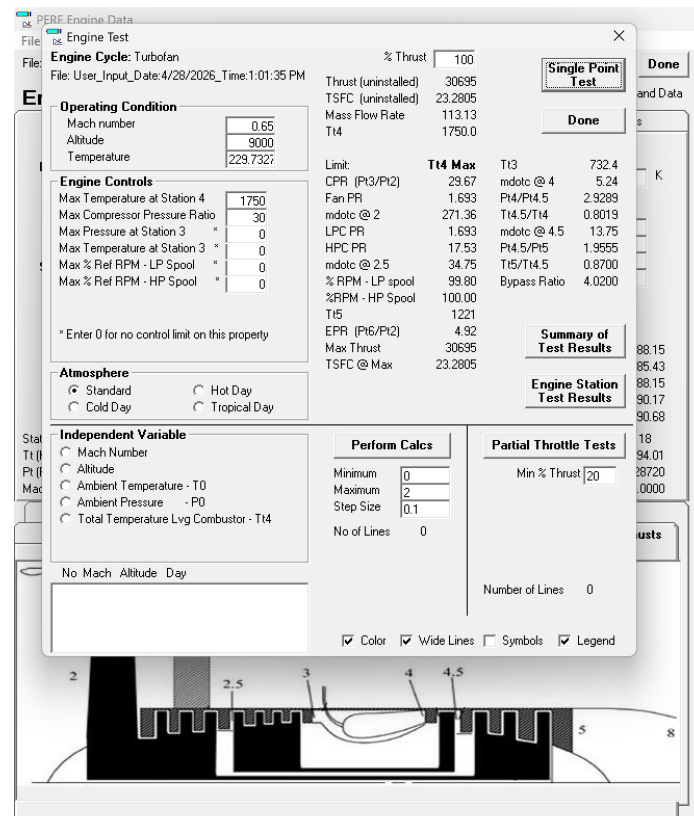


Figure 6. PERF result at mid-climb:  $M_0 = 0.65$ , Alt = 9,000 m.

Thrust/engine = 30,695 N; TSFC = 23.28 g/(kN · s); mission average required = 33,656 N. The slight shortfall at this mid-altitude point is offset by higher thrust at lower altitudes during the early climb.

### 5.3.3 Descent — $M_0 = 0.7$ , Altitude = 7,500 m

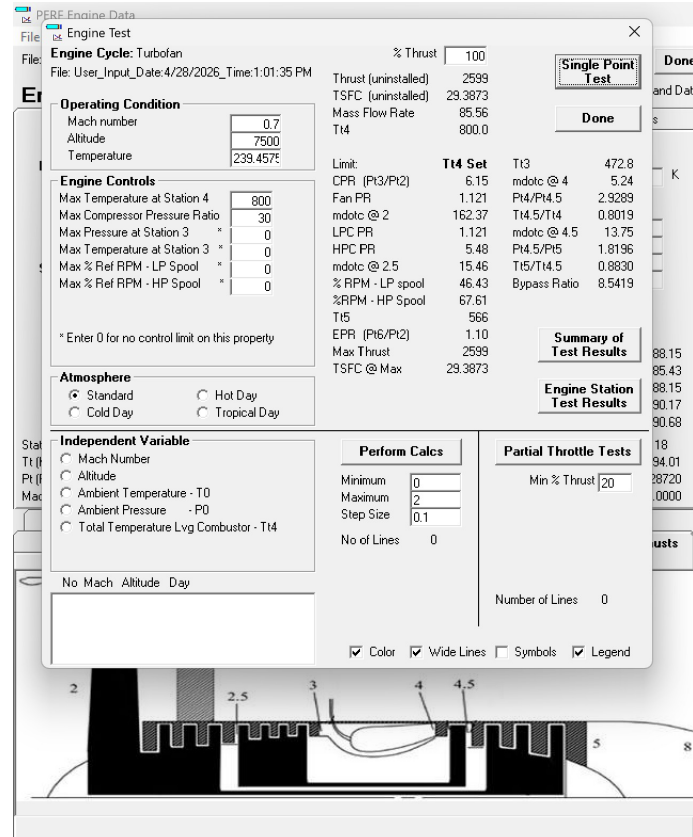


Figure 7. PERF result at descent:  $M_0 = 0.7$ , Alt = 7,500 m,  $T_{t4} = 800$  K (near-idle).

Near-idle thrust = 2,599 N/engine (vs. required 2,500 N); TSFC = 29.39 g/(kN · s); bypass ratio rises to 8.54 at off-design idle.

### 5.4 Engine Failure (OEI) Analysis

FAR 25.121 requires a minimum net climb gradient of 2.4% with one engine inoperative. At sea level, MTOW,  $V_{TO} = 77$  m/s:

$$C_D = 0.020 + \frac{1.099^2}{\pi \times 7.75 \times 0.85} = 0.0784; \quad D = 31,500 \text{ N} \quad (8)$$

$$\sin \theta_{OEI} = \frac{69283 - 31500}{435878} = 0.0867 \Rightarrow \mathbf{4.97\%} \quad (9)$$

Achieved OEI gradient **4.97%** vs. required **2.4%**  $\Rightarrow$  +107% margin.

## 6 Results and Discussions

### 6.1 Updated Fuel Burn with EPA-Derived TSFC

Table 8. Final summary of results — EPA-derived TSFC and updated fuel burn

| Phase        | Dist.(km)   | Thrust(kN) | TSFC  | Time(s)      | Fuel(kg)     |
|--------------|-------------|------------|-------|--------------|--------------|
| Taxi (out)   | 5.0         | 7.5        | 25.23 | 500          | 95           |
| Takeoff      | 20.0        | 132.2      | 18.16 | 53           | 127          |
| Climb        | 645.8       | 33.7       | 23.28 | 4306         | 3377         |
| Cruise       | 5558.4      | 21.8       | 25.23 | 22165        | 11880        |
| Descent      | 645.8       | 5.0        | 29.39 | 3588         | 527          |
| Land & Taxi  | 25.0        | 7.5        | 25.23 | 625          | 118          |
| <b>Total</b> | <b>6900</b> | —          | —     | <b>31237</b> | <b>16124</b> |

EPA total fuel burn is 16,124 kg vs. Part 2 assumed 15,735 kg — a difference of only +2.5%, confirming mission analysis accuracy. Fuel reserve =  $20049 - 16124 = 3,925$  kg (19.6%) — above the 10% minimum. Cruise dominates at 73.7% (11,880 kg) with EPA TSFC. All phases satisfy  $T_{\text{avail}} \geq T_{\text{req}}$ .

## 6.2 Ticket Cost Calculation

Fuel price: 1,50,000/kl; Jet-A density = 0.800 kg/l:

$$V_{\text{fuel}} = \frac{16124}{0.800} = 20,155 \text{ l} = 20.155 \text{ kl}; \quad C_{\text{fuel}} = 30,23,250$$

Fuel constitutes approximately 35% of total charter operating cost. The full breakdown:

**Table 9.** Charter cost breakdown — AMD to LHR, Gulfstream G650

| Cost Component        | Basis                 | Share(%)   | Amount ( )       |
|-----------------------|-----------------------|------------|------------------|
| Fuel                  | EPA burn + 1.5L/kl    | 35         | 30,23,250        |
| Crew (pilots + cabin) | Per-flight cost       | 15         | 12,95,250        |
| Navigation / ATC fees | International overfly | 10         | 8,63,500         |
| Airport handling      | AMD + LHR ground ops  | 8          | 6,90,800         |
| Maintenance & reserve | Engine cycle cost     | 17         | 14,67,450        |
| Insurance             | Per-flight liability  | 5          | 4,31,893         |
| Catering & misc.      | Passenger amenities   | 5          | 4,31,893         |
| Operator margin       | 10% markup            | 10         | 8,63,500         |
| <b>Total charter</b>  | —                     | <b>100</b> | <b>86,35,036</b> |

$$C_{\text{per seat}} = \frac{86,35,036}{13} \approx \boxed{6,64,234 \approx \text{USD } 8,000}$$

For reference, business-class commercial tickets on AMD–LHR cost approximately 1.8–2.5 lakhs and first class 4–5 lakhs. The private charter premium reflects exclusive aircraft use and the full operational overhead of a dedicated flight.

## 7 Conclusion

A complete engine design workflow was demonstrated for the Gulfstream G650 on the Ahmedabad–London route (6900 km). Mission analysis yielded a fuel burn of 15,735 kg with a 21.5% reserve margin. PCA identified Option B ( $\alpha = 4$ ,  $\pi_c = 30$ ,  $\pi_f = 1.7$ ) as the optimal design, offering a 30% specific thrust margin and  $\eta_O = 35.5\%$ . EPA using PERF V5.0 confirmed full mission feasibility: takeoff thrust margin +4.8%, OEI climb gradient 4.97%, and a fuel burn of 16,124 kg (+2.5% from Part 2). The estimated charter ticket cost is **6.64 lakhs/seat** ( $\approx$  USD 8,000).

## Acknowledgements

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## References

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